Do Learnable Connectivity Gates Identify Prunable Neurons? A Comparative Study of Structured Neural Network Pruning

Harnoor Singh
Undergraduate research project
https://github.com/Harnoor001/Self-Pruning-Network

Abstract—Self-pruning networks attach a learnable sigmoid gate to every connection and are often assumed to "learn" which neurons can be removed. We test this assumption in a controlled comparison. Starting from an existing self-pruning multilayer perceptron (MLP) for CIFAR-10, whose neuron score is the mean gate over a neuron's incoming connections, we compare that score with random, ℓ_1/ℓ_2 magnitude, activation, first-order Taylor and Network Slimming criteria under identical compact architectures, fine-tuning budgets and validation-based model selection, and we measure how well each score predicts the exact validation-loss increase caused by removing a single neuron. We show analytically that, when a gated layer is followed by batch normalization, the mean incoming gate of a neuron is invariant to the network function and therefore cannot be learned as an importance signal, and that an unnormalized ℓ_1 gate penalty under Adam moves all gates by the same amount. Over five seeds with 30-epoch sources, the mean-incoming-gate score was below random neuron selection at every pruning ratio and on every seed: after five fine-tuning epochs it reached 53.58% test accuracy against 57.95% for random at 60% of hidden units removed, and all 15 alternative criteria beat it (Holm-corrected paired t-test). Its scores were anti-correlated with single-neuron removal damage. Reading the same gates on the outgoing side, which batch normalization does not cancel, recovers the accuracy of magnitude criteria, and any positive gate penalty removes the failure, but no gate score exceeded the conventional criteria. A reduced-budget VGG-16-BN study on CPU (20-epoch sources, three fine-tuning epochs, three seeds) points the same way: gate-in was below random at 40 and 60% channel removal (83.6 vs. 85.0% at 60%), and at 60% every other criterion beat it on all three seeds, but at $n = 3$ only one of 24 differences survives Holm correction.

Index Terms—structured pruning, importance estimation, learnable gates, batch normalization, reproducibility

I. Introduction

Structured pruning removes whole neurons or channels and, unlike unstructured sparsity, yields physically smaller dense tensors whose speedup can be measured on commodity hardware [1], [2]. Its central decision is an importance criterion: which units can be removed with the least damage. Learnable gating is an attractive alternative to hand-designed criteria: if every connection carries a trainable multiplicative gate, gradient descent should reveal which parts of the network matter [3], [4].

This paper evaluates one concrete instance of that idea, an existing self-pruning MLP in which each weight W_{ij} is replaced by $W_{ij}\,\sigma(S_{ij})$ and neurons are ranked by their mean incoming gate. Rather than proposing a new algorithm, we ask five questions: whether the learned gate score identifies prunable neurons better than conventional criteria (RQ1, RQ2); how stable scores and decisions are across seeds (RQ3); how methods compare on measured parameters, MACs, size and latency (RQ4); and whether the estimated importance agrees with the actual damage of removing a neuron (RQ5).

Our contributions are: (i) a reproducible framework that applies ten criteria to the same source network and produces identical compact architectures for every criterion; (ii) an audit of the existing implementation, including a gate-folding error in its compact reconstruction (Section III); (iii) two elementary but consequential observations on gate learning with batch normalization and Adam (Section IV); and (iv) a reliability analysis that correlates each criterion with exact single-unit removal damage, following the methodology of Molchanov et al. [5].

II. Related Work

Magnitude criteria. Han et al. [6] prune small weights; Li et al. [1] rank filters by ℓ_1 norm. With batch normalization (BN) incoming-weight norms are scale-ambiguous [7], [8]. Data-driven criteria. Activation statistics [9], first-order Taylor expansions of the loss [10], [5], activation variance [11] and reconstruction of the next layer (ThiNet [12]) use calibration data. Recent work optimizes structures under measured latency constraints [13] or at finer-than-filter group granularity [14]. Learned scales and gates. Network Slimming [15] imposes an ℓ_1 penalty on BN scaling factors; L_0 regularization learns stochastic gates [3]; SNIP scores connections through multiplicative indicator variables [4]. SNACS aggregates connection-level scores between filters into structured decisions [16]; CoNNect regularizes for input–output connectivity [17]. Dependencies and dy-

namic pruning. DepGraph groups coupled parameters for arbitrary architectures [18]; Probe Pruning prunes dynamically per batch [19], a regime outside our static scope. Evaluation practice. Blalock et al. [20] and Liu et al. [21] show that unequal training budgets and single runs can invalidate pruning comparisons; our protocol follows their recommendations.

III. The Existing Self-Pruning Method

Architecture. The MLP maps a flattened 3072-dimensional image through hidden layers of widths 2048, 1024, 512 to 10 logits. Each hidden layer is Linear→BN→GELU→Dropout(0.3).

Gates. Every linear layer l has weights $W_l \in \mathbb{R}^{n_l \times n_{l-1}}$ (row j = incoming connections of neuron j), an ungated bias b_l and gate scores S_l of the same shape, initialized to 2 ($\sigma(2) \approx 0.881$):

$$h_l = (W_l \odot \sigma(S_l))\, x_{l-1} + b_l. \tag{1}$$

Objective. With label-smoothed cross-entropy $\mathcal{L}_{\mathrm{CE}}$,

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \sum_l \sum_{i,j} \sigma(S_{l,ji}), \tag{2}$$

an unnormalized ℓ_1 penalty over all 8.9M gates, optimized with AdamW over all parameters (gate scores included) for five epochs.

Structured pruning. The importance of hidden neuron j in layer l is the mean of its incoming gates,

$$I^{\mathrm{in}}_{l,j} = \frac{1}{n_{l-1}} \sum_i \sigma(S_{l,ji}), \tag{3}$$

and each hidden layer keeps its $n_l - \lfloor r\, n_l \rfloor$ highest-scoring neurons for a removal ratio r. Rows of W_l, b_l and the BN parameters and the corresponding columns of W_{l+1} are sliced into a smaller dense network, which is then fine-tuned.

Audit. The reconstruction copied W_l instead of the effective weight $W_l \odot \sigma(S_l)$, so the compact model differed from its source even at $r = 0$ (validation accuracy 51.74% → 50.70% on the seed-42 checkpoint; research/INITIAL_AUDIT.md). We fold gates before slicing in all experiments below; the legacy behavior is retained only for reproducing old artifacts.

IV. Two Properties of Gate Learning

Batch-norm invariance of incoming gates. Let neuron j's pre-activation be $h_j = \sum_i W_{ji} g_{ji} x_i + b_j$ with gates $g_{ji} = \sigma(S_{ji})$, followed by BN. Replacing $g_{j\cdot}$ by $c\, g_{j\cdot}$ for any $c > 0$ scales the centred pre-activation $h_j - \mathbb{E}_B[h_j]$ by c (the bias cancels in the centring) and its batch standard deviation by c, so the BN output is unchanged up to the ϵ term. Hence the overall level of a neuron's incoming gates—exactly what I^{in} measures—does not affect the training-mode function and receives no consistent gradient. The argument is a direct consequence of BN's scale invariance [8] (cf. [7] for weight norms); a unit test in our repository verifies it numerically. Outgoing gates g_{kj} in layer $l+1$ scale neuron j's contribution to every downstream pre-activation and are not cancelled. This motivates two variants evaluated separately from the existing method: the mean outgoing gate $I^{\mathrm{out}}_{l,j} = \mathrm{mean}_k\, \sigma(S_{l+1,kj})$ and their mean-normalized sum $I^{\mathrm{comb}} = I^{\mathrm{in}}/\overline{I^{\mathrm{in}}} + I^{\mathrm{out}}/\overline{I^{\mathrm{out}}}$.

Uniform drift under Adam. The penalty gradient $\partial(\lambda\sigma(S))/\partial S = \lambda\sigma'(S)$ is identical for gates with equal scores. When it dominates the data gradient, Adam's per-coordinate normalization makes every gate take the same step. In the committed $\lambda = 10^{-3}$ checkpoint, all hidden-layer gates lie in $[0.7417, 0.7431]$ after training, consistent with this mechanism; with $\lambda = 0$ the neuron scores span only $[0.8798, 0.8808]$ in the first layer. These are the gates whose rankings Method 6 relies on.

Relation to Taylor saliency. For a gated connection, $\partial\mathcal{L}/\partial g = W\, \partial\mathcal{L}/\partial(Wg)$, so $g\, \partial\mathcal{L}/\partial g$ is the first-order Taylor change of the loss when the connection is removed; gate training integrates (a sign-normalized version of) this SNIP-like saliency [4], [5].

V. Experimental Setup

Data. CIFAR-10 [22]: 45,000/5,000 train/validation and the 10,000-image test set, used only for final reporting. The research protocol uses a split fixed across seeds; the verification experiment reproduces each legacy checkpoint's own seed-dependent split exactly so that validation images were never trained on. Activation and Taylor scores use the first 2,048 training images without augmentation.

Sources. For each seed, the dense source is trained with $\mathcal{L}_{\mathrm{CE}}$ only, the gated source with the existing objective, and the Slimming source with $\mathcal{L}_{\mathrm{CE}} + 10^{-4} \sum |\gamma|$. All three share the same initialization for a given seed and are trained for 30 epochs (AdamW, learning rate 8×10^{-4}, cosine schedule, label smoothing 0.05); the best-validation epoch is kept. The gated source uses $\lambda = 0$, the setting of the existing checkpoints; other penalties are studied in ablation A. Seeds are 42, 123, 2024, 7 and 21. Compact models are fine-tuned for 5 epochs (AdamW, 2×10^{-4}, cosine). In the verification experiment (Section VII-D) the dense and gated sources are instead the committed 5-epoch checkpoints, fine-tuned for one epoch.

Common pruning protocol. All post-training criteria score the same gated source (with gates folded into the weights), and are repeated on the dense source to isolate the training objective. Uniform allocation removes $\lfloor r n_l \rfloor$ units per layer, $r \in \{0.2, 0.4, 0.6\}$, so every criterion produces the same architecture. Every compact model is fine-tuned with the identical optimizer, schedule, budget and data order; the checkpoint with the best validation accuracy (including the un-fine-tuned model) is reported. An unpruned + fine-tuning control receives the same budget so that extra training is not credited to pruning [21].

Reliability. For 64 randomly sampled units per layer we zero the unit's output (exactly equivalent to removing it, in evaluation mode; verified by tests), record the increase

of validation loss, and compute Spearman's ρ between this damage and each criterion's score, per layer and averaged over layers.

Statistics. We report mean ± sample standard deviation over seeds and 95% t-intervals. Paired comparisons against the existing method use matched seeds with a paired t-test and an exact Wilcoxon signed-rank test, Holm-corrected within each ratio [23]. With $n = 3$ seeds the smallest attainable two-sided Wilcoxon p is 0.25; we therefore treat all differences as descriptive unless the Holm-adjusted t-test gives $p < 0.05$, and we state the normality assumption explicitly.

VI. Baseline Methods

Let $W_l[j]$ be neuron j's incoming weights and $W_{l+1}[:, j]$ its outgoing weights (all kernel elements for convolutions), a_j its post-activation output, and $\bar{\cdot}$ a layer mean.

$$I_j^{\ell_p} = \frac{\|W_l[j]\|_p}{\overline{\|W_l[\cdot]\|_p}} + \frac{\|W_{l+1}[:, j]\|_p}{\overline{\|W_{l+1}[:, \cdot]\|_p}}, \quad p \in \{1, 2\}, \tag{4}$$

$$I_j^{\text{act}} = \mathbb{E}_x \, \text{mean}_{\text{sp}} |a_j(x)|, \tag{5}$$

$$I_j^{\text{Taylor}} = \mathbb{E}_x \Big| \textstyle\sum_{\text{sp}} a_j(x) \, \partial \mathcal{L}(x) / \partial a_j(x) \Big|, \tag{6}$$

$$I_j^{\text{slim}} = |\gamma_{l,j}| \text{ (on the Slimming source)}, \tag{7}$$

plus random scores and $|\gamma|$ on non-Slimming sources as a diagnostic. Slimming is also reported with its original global ranking at the same total unit budget.

VII. Results and Discussion

Executed experiments. The main study follows the research protocol (configs/mlp.yaml): 30-epoch dense, gated ($\lambda = 0$) and Slimming sources for five seeds, 5 fine-tuning epochs for every compact model, and ablations A and C on three seeds. All of it ran on CPU (23 h). The sources reach 60.06 ± 0.17 (dense), 60.14 ± 0.16 (gated) and 59.51 ± 0.08% (Slimming) test accuracy. Section VII-D summarizes the earlier verification on the original 5-epoch checkpoints; Section VII-E reports a reduced-budget VGG-16-BN study, also on CPU. The full-budget VGG-16-BN protocol was not executed (no GPU).

A. Pruning accuracy (RQ1, RQ2)

Table I and Fig. 1 show test accuracy after fine-tuning. The existing gate-in score gives the lowest accuracy of all criteria at every ratio: 58.26, 56.29 and 53.58% at 20, 40 and 60% removal, against 59.98, 59.22 and 57.95% for random selection on the same source. Before fine-tuning the gap is larger: 44.8/30.5/18.9% for gate-in against 57.4/46.9/24.0% for random (Fig. 2). All 15 alternative criteria beat gate-in on all five seeds at every ratio, by +1.2 to +2.2 pp at 20%, +2.9 to +3.7 pp at 40% and +4.4 to +5.3 pp at 60%. The Holm-corrected paired t-test supports every one of these 45 differences at $p < 0.05$ (largest adjusted p: 0.0046 at 20%). With five pairs the exact Wilcoxon test cannot fall below $p = 0.0625$ and reaches that floor for every comparison; Shapiro–Wilk rejects normality of the differences for 4 of the 45 comparisons, all at 20%.

TABLE I
mlp: test accuracy (%, mean ± sample SD over 5 seeds) after 5 fine-tuning epoch(s). Criteria score the gated source except Net. Slimming (own training objective); identical compact architectures per column.

Method	Hidden units removed 20%	40%	60%
Gate-in (existing)	58.26±0.24	56.29±0.16	53.58±0.28
Taylor	60.43±0.24	59.98±0.13	58.82±0.33
Activation	60.17±0.40	59.60±0.41	58.37±0.18
L1 in+out	60.39±0.09	59.96±0.26	58.82±0.21
L2 in+out	60.26±0.42	60.04±0.14	58.63±0.42
Net. Slimming	59.41±0.23	59.27±0.29	58.57±0.19
Gate-combined (V2)	60.26±0.17	59.68±0.28	58.66±0.21
Gate-out (V1)	60.16±0.21	59.87±0.21	58.72±0.08
Random	59.98±0.24	59.22±0.22	57.95±0.07
Unpruned gated + FT		60.46±0.20	
Unpruned dense + FT		60.27±0.23	
Unpruned slimming + FT		59.76±0.24	

The separately implemented gate variants, mean outgoing gate (V1) and the combined score (V2), reach the level of the magnitude criteria (e.g. 58.72 and 58.66% at 60%, against 58.82% for ℓ_1 and Taylor). This is consistent with Section IV: BN cancels a neuron's incoming-gate level but not its outgoing gates. In layer 1 the gate-in scores have a within-layer coefficient of variation of $\approx 9 \times 10^{-4}$ on every seed, against 0.11–0.12 for ℓ_1 and 0.57–0.59 for Taylor. We did not test differences among the other criteria and claim no ranking among them. Descriptively, Taylor, ℓ_1, ℓ_2 and $|\gamma|$ lie within 0.35 pp of each other, and at 20% Taylor and ℓ_1 are within 0.1 pp of the unpruned + fine-tuning control (60.46%). The same criterion on the dense source gives nearly identical accuracy (ℓ_1 at 60%: 58.78 dense vs. 58.82% gated), so gate training at $\lambda = 0$ did not make the network more prunable. Network Slimming retains the most accuracy before fine-tuning at 40–60%, but after fine-tuning it ends 0.25–1.0 pp below Taylor on the gated source; its own source is 0.6 pp weaker, so criterion and source are confounded. Global allocation for Slimming yields different, smaller architectures (−32/−59/−78% parameters) and is not an equal-cost comparison.

B. Agreement with single-unit removal damage (RQ5)

Fig. 3 reports Spearman's ρ between each score and the validation-loss increase caused by removing one unit (64 units per layer, 3 layers, 5 seeds). Gate-in is the lowest criterion and negatively correlated with damage ($\rho = -0.161 \pm 0.088$; per layer −0.08, −0.21, −0.19), below random (-0.043 ± 0.077). On the dense and gated sources every correlation is weak ($|\rho| \leq 0.17$): removing one unit changes the loss by 1.7×10^{-4} nats on average, and 23–24% of removals decrease it. On the Slimming source all non-random criteria correlate more strongly with damage (0.22–0.32), and fewer removals reduce the loss (20%). Taylor and activation scores are stable under resampling

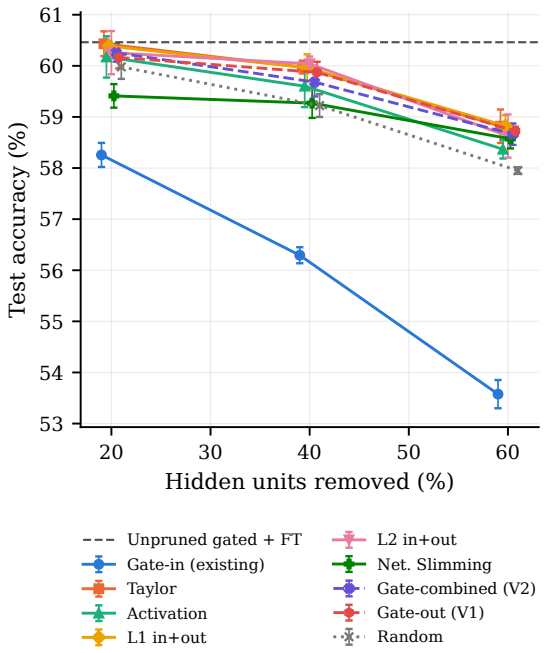


Fig. 1. Test accuracy after five fine-tuning epochs (mean ± SD, 5 seeds; markers offset horizontally for legibility). Dashed: unpruned gated model with the same fine-tuning budget.

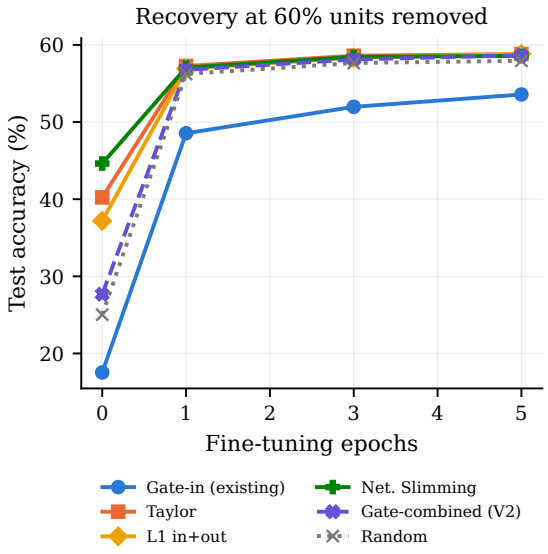


Fig. 2. Accuracy immediately after pruning (dotted) and after fine-tuning (solid), mean over 5 seeds.

of the calibration set (Spearman 0.96–0.99 between two disjoint subsets), so their weak reliability is not estimation noise. Single-unit damage is not additive over the large groups removed in practice, so RQ5 and RQ1 measure different properties.

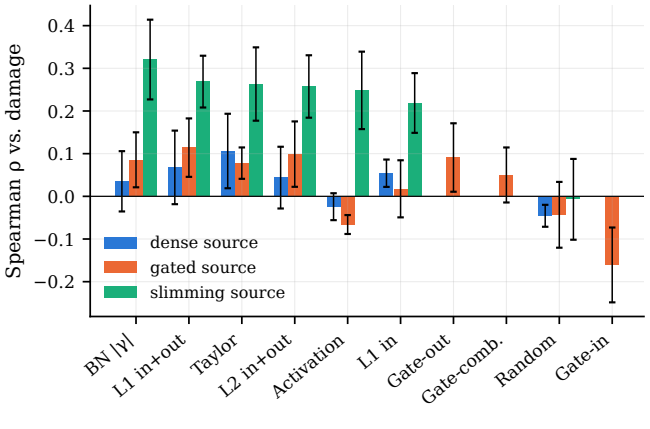


Fig. 3. Spearman ρ between importance score and single-unit removal damage (mean over layers, then mean ± SD over 5 seeds). Gate criteria exist only for the gated source.

C. Measured efficiency (RQ4)

Uniform allocation gives identical architectures for every criterion. Deployable parameters fall from 8.93 M to 6.73, 4.73 and 2.95 M (−24.6, −47.0, −67.0%), with MACs and serialized size reduced proportionally; these are physical reductions of dense tensors. Median single-thread CPU latency at batch size 64 falls from 11.95 ms to 9.47, 6.56 and 4.43 ms. Batch-1 latency also fell in this run (1.17 → 0.84, 0.52, 0.32 ms), but in the verification experiment the 20% model measured slower than dense at batch size 1, so we claim speed-ups only at batch size 64. The gated model as trained, which recomputes $W \odot \sigma(S)$ in every forward pass, takes 12.98 ms at batch size 1 ($\approx 11\times$ the dense model); gates must be folded into the weights before deployment. No GPU measurements were made.

D. Verification on the original checkpoints

Before the main study we evaluated the committed 5-epoch checkpoints of the existing repository (3 seeds, ≈ 52% test accuracy, 1 fine-tuning epoch, each seed on its own original split). The outcome was the same: gate-in was below random at every ratio (47.05 vs. 50.14% at 60%), all 13 alternatives beat it on every seed, and its scores were negatively correlated with removal damage ($\rho = -0.119 \pm 0.039$). Full tables are in research/results_analysis.md, Part II.

E. Reduced-budget VGG-16-BN on CIFAR-10

Setup and deviations. VGG-16-BN with a gate on every convolution weight, the same fixed split and the selection rule above (configs/vgg_cifar10_cpu.yaml). Dense, gated ($\lambda = 0$) and Slimming sources are trained with SGD (Nesterov, learning rate 0.05, weight decay 5×10^{-4}, none on gates, cosine schedule). Because only a CPU was available, the budget is reduced relative to the specified protocol: 20 instead of 160 source epochs, 3 instead of 20 fine-tuning epochs (SGD, 0.01), seeds 42, 123 and 2024

TABLE II
vgg16_cpu: test accuracy (%, mean ± sample SD over 3 seeds) after 3 fine-tuning epoch(s). Criteria score the gated source except Net. Slimming (own training objective); identical compact architectures per column.

Method	Hidden units removed 20%	40%	60%
Gate-in (existing)	89.45±0.42	87.32±0.17	83.60±0.84
Taylor	90.03±0.35	88.94±0.12	86.78±0.13
L1 in+out	89.68±0.22	88.86±0.21	86.78±0.45
Net. Slimming	89.34±0.20	88.54±0.02	86.42±0.23
Gate-combined (V2)	89.94±0.28	88.84±0.35	86.27±0.10
Gate-out (V1)	89.56±0.52	88.42±0.24	86.54±0.21
Random	89.31±0.39	87.59±0.43	85.02±0.74
Unpruned gated + FT		90.41±0.36	
Unpruned slimming + FT		89.76±0.16	
Unpruned dense + FT		89.55±0.42	

instead of five, 9 of 16 methods, 16 instead of 64 reliability units per layer, and no ablations. The run took about 20 h per seed. The sources reach 89.55 ± 0.42 (dense), 90.41 ± 0.36 (gated) and 89.76 ± 0.16% (Slimming) test accuracy and were still improving at epoch 20. These results are not results of the full protocol.

Accuracy. Table II and Fig. 4 show the same direction as the MLP. Gate-in is the lowest criterion at 40 and 60% channel removal (87.32 and 83.60%), below random (87.59 and 85.02%), while Taylor and ℓ_1 reach 88.9 and 86.8%. At 20% all criteria lie within 0.9 pp and gate-in (89.45%) is indistinguishable from random (89.31%). Before fine-tuning, removing 20% of channels by gate-in leaves the network at chance (10.3%), against 25.3% for random and 65–70% for ℓ_1 and Slimming. Every non-random method beats gate-in on all three seeds at 40% (by 1.1–1.6 pp) and every method at 60% (by 2.7–3.2 pp; random by 1.4 pp). With $n = 3$, however, only Taylor at 40% survives Holm correction (+1.62 pp, 95% CI [1.28, 1.97], adjusted $p = 0.019$); at 60% the smallest adjusted p is 0.064. The VGG evidence is therefore consistent with, but much weaker than, the MLP result. Parameters fall by 35.8, 63.8 and 83.9% (14.7 M→ 2.4 M at 60%); measured CPU latency at batch size 64 falls from 499 to 354, 297 and 147 ms.

Gate statistics and reliability. With $\lambda = 0$ and SGD the gates barely moved: every layer's mean gate is $\sigma(2) = 0.881$ to three decimals, and the within-layer coefficient of variation of the gate-in score is about 6×10^{-8} (gate-out $\approx 2 \times 10^{-7}$), against 0.07 for ℓ_1 and 0.4–0.5 for Taylor, so the rankings rest on displacements near float32 resolution. They are nevertheless not neutral: gate-in is worse than random, whereas gate-out ranks channels about as well as ℓ_1, the same incoming/outgoing contrast as in the MLP. Spearman correlations with single-channel removal damage are weak for every criterion (0.10–0.20 for non-random criteria on the gated source) and lowest for gate-in (0.033 ± 0.053), which is strongly negative only in the first convolution ($\rho = -0.67$). 42% of single-channel removals lower the validation loss, so per-layer correlations in the deeper layers are dominated by noise. Whether a positive gate penalty would remove the VGG deficit, as it did for the MLP, was not tested.

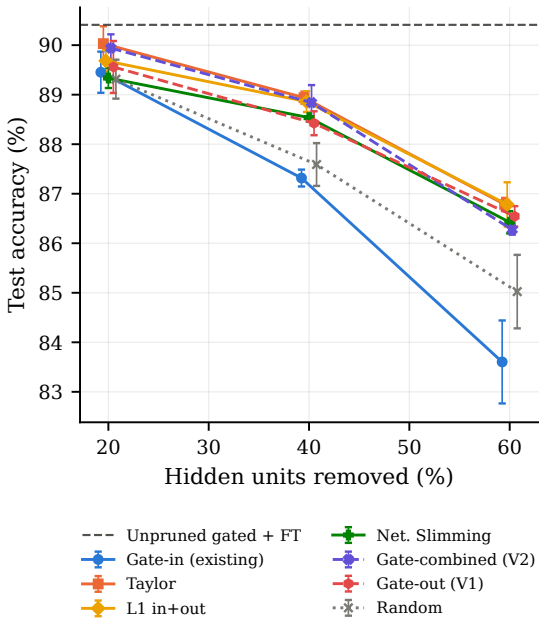


Fig. 4. Reduced-budget VGG-16-BN (20-epoch sources): test accuracy after three fine-tuning epochs (mean ± SD, 3 seeds). Dashed: unpruned gated model with the same fine-tuning budget.

VIII. Ablation Studies

The framework defines five ablations: A, gate penalty strength; B, gate aggregation (incoming, outgoing, combined); C, fine-tuning budget; D, pruning ratio; E, seeds. B, D and E are part of the main study above. A and C were run on seeds 42, 123 and 2024 and are descriptive; no hypothesis tests were run on them.

Ablation C (fine-tuning budget). Table III varies the budget over $\{0, 1, 3, 5\}$ epochs. Fine-tuning shrinks but does not close the gate-in deficit: at 60% removal its gap to random is 7.5, 7.7, 5.7 and 4.4 pp after 0, 1, 3 and 5 epochs. The other methods gain at most 0.55 pp from 3 to 5 epochs, so the main budget is close to saturation for them. Without fine-tuning, Network Slimming is best at 40 and 60%, but not after five epochs. Method rankings therefore depend on the fine-tuning budget, which is why the budget is identical for every method.

Ablation A (gate penalty). Table IV and Fig. 5 train separate gated sources with the existing penalty at $\lambda \in \{10^{-6}, \ldots, 10^{-3}\}$ (sum) and with a mean-normalized penalty ($\lambda \in \{0.1, 1\}$), and prune 40% of units. The existing method's formulation is unchanged; only λ varies. Three observations follow. (i) The gate-in failure is specific to $\lambda = 0$: with every positive penalty, gate-in is within 0.25 pp of gate-out and gate-combined and 0.36–0.98 pp above random. The penalty creates the spread the score needs (within-layer SD of the gate-in score $\leq 8 \times 10^{-4}$

TABLE III
Ablation C: mean test accuracy (%) versus fine-tuning epochs. The primary budget uses all seeds; the other budgets use the seeds on which ablation C was run (see findings.md for n per cell).

Method	Removed	0 ep	1 ep	3 ep	5 ep
Gate-combined (V2)	20%	55.40	59.09	60.38	60.26
Gate-combined (V2)	40%	44.87	58.41	59.35	59.68
Gate-combined (V2)	60%	27.67	56.77	58.11	58.66
Gate-in (existing)	20%	45.10	56.38	57.84	58.26
Gate-in (existing)	40%	30.54	53.05	55.50	56.29
Gate-in (existing)	60%	17.53	48.55	51.97	53.58
L1 in+out	20%	57.40	59.24	60.34	60.39
L1 in+out	40%	50.11	58.63	59.67	59.96
L1 in+out	60%	37.19	56.92	58.34	58.82
Random	20%	57.67	59.17	59.99	59.98
Random	40%	47.76	57.83	59.12	59.22
Random	60%	25.03	56.23	57.66	57.95
Net. Slimming	20%	58.67	58.67	59.30	59.41
Net. Slimming	40%	55.70	58.40	59.10	59.27
Net. Slimming	60%	44.62	57.06	58.46	58.57
Taylor	20%	58.68	59.43	60.25	60.43
Taylor	40%	54.01	58.84	59.77	59.98
Taylor	60%	40.23	57.27	58.60	58.82

TABLE IV
Ablation A: test accuracy (%, mean over 3 seeds) when the gated source is trained with different gate penalties.

Penalty	Source	Gate-comb.	Gate-in	Gate-out	Random
λ=0	60.25	59.63	56.30	59.84	59.27
λ=1e-06 (sum)	60.32	59.76	59.85	59.88	59.49
λ=1e-05 (sum)	60.09	59.18	59.30	59.35	58.68
λ=0.0001 (sum)	60.20	58.33	58.31	58.55	57.62
λ=0.001 (sum)	60.05	57.93	57.90	58.04	57.52
λ=0.1 (mean)	60.23	59.91	59.92	59.91	58.94
λ=1 (mean)	60.28	59.86	59.69	59.81	59.09

at $\lambda = 0$, 0.025–0.12 at $\lambda = 10^{-6}$). (ii) Source accuracy is unaffected (60.05–60.32%) even at $\lambda = 10^{-3}$, where all connection gates have collapsed to $\sigma(S) \approx 0.087$ with a within-layer SD below 2.5×10^{-4}. Batch normalization absorbs the uniform rescaling, a direct observation of both properties in Section IV. (iii) Stronger sum penalties lower post-pruning accuracy for all criteria, random included (59.49 → 57.52%), so they make the network harder to prune at this budget rather than making the gates more informative. The mean-normalized penalties show no such drop.

IX. Limitations

- Executed scope. The MLP study (five seeds), its ablations A and C (three seeds) and the verification on the original checkpoints were executed on CPU. The VGG-16-BN study was executed only at a reduced budget (Section VII-E: 20 instead of 160 source epochs, 3 instead of 20 fine-tuning epochs, 3 instead of 5 seeds, 9 of 16 methods, no ablations); the full protocol is implemented but was not run.
- Model quality. The MLP sources reach about 60% test accuracy on raw pixels; conclusions may differ for stronger models.

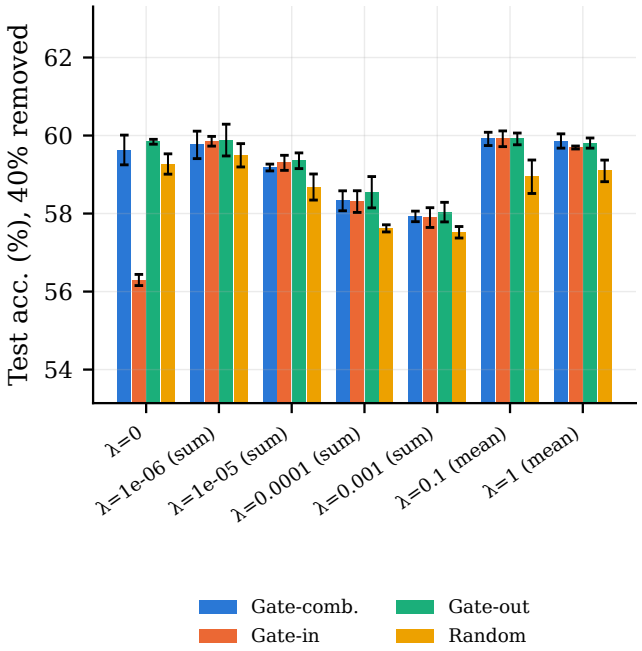


Fig. 5. Ablation A: test accuracy after removing 40% of hidden units from gated sources trained with different penalties (mean ± SD, 3 seeds, 5 fine-tuning epochs).

- Statistical power. With five seeds the exact Wilcoxon test cannot reach $p < 0.0625$, so significance rests on the paired t-test, whose normality assumption is questionable for 4 of 45 comparisons. Ablations A and C (three seeds) are descriptive. The VGG comparisons use three seeds (Wilcoxon floor $p = 0.25$); only one survives Holm correction, so they are largely descriptive.
- Confounds. Network Slimming is evaluated on its own source, which is 0.6 pp weaker; global allocation changes the architecture and is not an equal-cost comparison.
- Hardware. Latency was measured on one laptop CPU; no GPU was available, so no GPU latency or memory figures are reported.
- Architecture. The MLP on raw pixels is not representative of modern vision models. The CNN evidence comes from one reduced-budget VGG-16-BN study whose gated source was trained with $\lambda = 0$ only; the gate-penalty ablation was not repeated for VGG.

X. Conclusion

We asked whether the learnable connection gates of an existing self-pruning MLP identify prunable neurons. Every criterion used the same compact architectures, fine-tuning budget and validation-based selection, over five seeds with 30-epoch sources. In the setting the existing method ships with ($\lambda = 0$) the answer is no: its neuron score, the mean incoming gate, gave lower accuracy than

random neuron selection at every pruning ratio and on every seed, by up to 4.4 pp after fine-tuning, and its scores were anti-correlated with the measured damage of removing a neuron. The same result held on the original 5-epoch checkpoints. This matches the analysis in Section IV: behind batch normalization, the level of a neuron's incoming gates does not affect the network function.

Three outcomes are constructive. First, reading the same gates on the outgoing side, which BN does not cancel, recovers the accuracy of standard magnitude criteria. Second, any positive gate penalty we tested removes the failure: gate-in then matches the other gate scores and exceeds random selection (descriptively, three seeds). Even so, in the main study no gate score exceeded the conventional criteria, and stronger sum penalties made the networks harder to prune for every criterion. Third, the audit found and corrected a gate-folding error in the original compact reconstruction. The results do not show that gate-based criteria are better than conventional ones, and they establish no ranking among the conventional criteria, which were not tested against each other.

A reduced-budget VGG-16-BN study (20-epoch sources, three fine-tuning epochs, three seeds) points in the same direction: gate-in fell below random at 40 and 60% channel removal and lost to every other criterion on every seed at 60%. With three seeds, however, only one comparison survives multiple-testing correction, so this is supporting rather than independent evidence for convolutional networks. The main conclusions rest on one MLP architecture on raw CIFAR-10 pixels, CPU-only measurements and three seeds for the ablations. The full VGG-16-BN protocol (160-epoch sources, 20 fine-tuning epochs, five seeds, gate-penalty ablation) on a GPU is the necessary next step before the findings can be generalized.

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